

Example 1

CN PRACTICE

1

Some transmitter sends six packets 0, 1, 2, 3, 4 and 5. It receives ACK with ACK no. = 3, how can you express it considering GBN or SR?

SOL :-

Considering GBN :-

packets 0, 1 & 2 have been received successfully & receiver is expecting 3.

Considering SR :-

It shows, packet 3 has been received successfully, the ACK does not say anything about other packets.

TRANSPORT LAYER

Example 2 :-

A TCP connecting is transferring a file of 5K bytes. The first byte is numbered 10001.

What are the sequence nos. for each segment if data are sent in five segments, each carrying 1000 bytes? (Done in class).

Example 3.

(2)

If MSS = 10KB and RTT = 100ms, calculate the throughput.

SOL :-

$$\text{Throughput} = \frac{W_{\max} (0.75)}{\text{RTT}}$$

$$W_{\max} = \frac{(10 + 12 + 10 + 8 + 8)}{5} = 9.66 \text{ MSS}$$

$W_{\max}$ is the average of window sizes when the congestion occurs (given in exam).

So,

$$\text{Throughput} = \frac{0.75 \times 9.66 \times 10\text{KB}}{100\text{ms}}$$

$$= \underline{7.2 \text{ Mbps}} \text{ Ans.}$$

in bps.

Example 4.

Consider the data :

RTT	Cwnd	Status
1	1	initial
2	2	double
3	4	"
4	8	"
5	16	"

Given: cwndo = 1 MSS and ssthreshold = 16 MSS, calculate RTT before congestion avoidance.

SOL -

It takes 4 RTTs before congestion avoidance.

Exaple 6:

Why does the last fragment in IP fragmentation often have a data size smaller than the MTU?

Sol:-

B/c the total datagram size is not an exact multiple of the data portion that fits in the maximum transmission unit.

Exaple 7:-

A 2400 byte datagram passes thru a router with MTUs 1500 bytes. Find total no. of fragments, consider header size = 20 bytes.

SOL:-

4

Data per fragment = $1500 - 20 = 1480$ bytes

Remaining data = $2400 - 1480 = 920$.

So, total two fragments are required.

Example 8

If a TCP transmission has Estimated RTT = 900ms , DevRTT = 100ms and the new SampleRTT = 1200ms :

Compute the new TimeoutInterval using $\alpha = 0.125$ and $\beta = 0.25$.

SOL:-

New Estimated RTT:

$$= (1 - 0.125) \times 900 + 0.125 \times 1200 = 937.5\text{ms}$$

New DevRTT :-

$$= (1 - 0.25) \times 100 + 0.25 \times |1200 - 937.5|$$
$$= 140.6\text{ms}$$

TimeoutInterval :-

$$= 937.5 + 4(140.6) = 1.55\text{ s Ans.}$$